SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01 **Total Marks:100**

Annexure A

CASE STUDY

Code of Conduct:

I G SAI TEJA 192472137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 1: E-Commerce Customer Analysis
Scenario:
An e-commerce company has collected customer transaction data, including purchase amount, number of orders, browsing time, and customer ratings. The company wants to understand customer purchasing behavior and improve its marketing strategies.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize customer transaction data?
B. Discuss how unusually high or low purchase amounts can affect the analysis and describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether premium members spend significantly more than regular customers? Formulate the hypotheses and explain the analysis process.

1. Introduction

E-commerce platforms generate a large amount of customer data through online purchases and browsing activities. This data can provide valuable information about customer purchasing behavior, spending patterns, and preferences. By analyzing this information, an e-commerce company can understand its customers better and make informed business decisions.

In this case study, the company has customer-related data such as purchase amount, number of orders, browsing time, and customer ratings. Analyzing these variables helps the company understand how much customers spend, how frequently they place orders, and how their online activity is related to their purchasing behavior.

Statistical techniques are used to convert this raw customer data into meaningful information. Descriptive statistics such as mean, median, standard deviation, and quartiles are used to summarize the purchase amounts and understand their distribution. Outlier detection is used to identify unusually high or low purchase values that may affect the overall analysis.

The company also wants to determine whether premium members spend more than regular customers. For this purpose, hypothesis testing can be used to statistically compare the average purchase amounts of the two customer groups. This helps determine whether the observed difference is significant or could have occurred by chance.

Therefore, this analysis combines descriptive statistics, outlier detection, and hypothesis testing to understand customer purchasing behavior and support better marketing and business decisions. The case study specifically focuses on summarizing transaction data, identifying unusual purchase amounts, and testing whether premium customers have significantly higher spending than regular customers.

2. Problem Statement

The e-commerce company wants to analyze its customer transaction data to understand purchasing behavior and spending patterns. The available data contains information such as purchase amount, number of orders, browsing time, and customer ratings.

The problem is divided into three main areas:

A. Descriptive Statistical Analysis

The company needs to summarize customer purchase amounts using statistical measures such as:

- **Mean** – to determine the average purchase amount.
- **Median** – to identify the middle purchase value.
- **Standard deviation** – to measure how much purchase amounts vary from the average.
- **Quartiles** – to understand the distribution of customer spending.

B. Outlier Detection

The company needs to identify unusually high or low purchase amounts. Such values may affect the mean and standard deviation and could lead to incorrect business conclusions.

The Interquartile Range (IQR) method is used to detect these outliers by calculating the lower and upper limits of the normal range.

C. Hypothesis Testing

The company also wants to determine whether premium members spend more than regular customers.

The analysis will compare the average purchase amounts of the two groups using hypothesis testing.

- **Null Hypothesis (H_0):** Premium and regular customers have the same average spending.
- **Alternative Hypothesis (H_1):** Premium customers spend more than regular customers.

The final objective is to use statistical analysis to obtain meaningful insights into customer behavior and help the company make better marketing and business decisions.

3. Procedure

1. Collect the customer transaction data containing purchase amount, number of orders, browsing time, and customer ratings.
2. Load and organize the data into a structured dataset using Python and Pandas.
3. Select the purchase amount data for statistical analysis.
4. Calculate descriptive statistics:
    - Mean
    - Median
    - Standard deviation
    - First quartile (Q1)
    - Third quartile (Q3)
5. Calculate the Interquartile Range (IQR) using:

$$IQR = Q3 - Q1$$

6. Detect outliers using the IQR method by calculating:

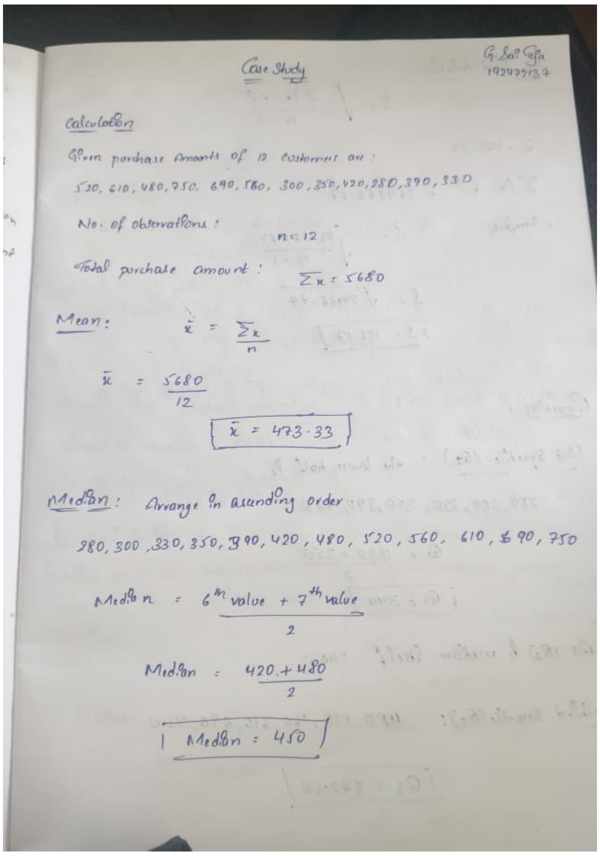
$$\text{Lower Bound} = Q1 - 1.5(IQR)$$
$$\text{Upper Bound} = Q3 + 1.5(IQR)$$

7. Separate the customers into two groups:
    - Premium customers
    - Regular customers
8. Calculate the average purchase amount for both groups.
9. Formulate the hypotheses:
    - H_0: Premium and regular customers have the same average spending.
    - H_1: Premium customers spend more than regular customers.
10. Perform an independent two-sample t-test and obtain the t-statistic and p-value.
11. Compare the p-value with the significance level (0.05) to decide whether to reject or fail to reject H_0.
12. Interpret the results and identify the customer spending pattern to support marketing decisions.

4.Dataset – E-Commerce Customer Analysis

Customer	Type	Purchase Amount	Orders	Browsing Time	Rating
C1	Premium	520	5	45	4.5
C2	Premium	610	6	52	4.8
C3	Premium	480	4	38	4.2
C4	Premium	750	7	60	4.9
C5	Premium	690	6	55	4.6
C6	Premium	560	5	48	4.4
C7	Regular	300	3	30	3.8
C8	Regular	350	4	35	4.0
C9	Regular	420	4	40	4.1
C10	Regular	280	2	25	3.5
C11	Regular	390	3	33	3.9
C12	Regular	330	3	28	3.7

Calculations:



Case Study

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192472137

Calculation

Given purchase amounts of 12 customers are :

520, 610, 480, 750, 690, 560, 300, 350, 420, 280, 390, 330

No. of observations :

$n = 12$

Total purchase amount : $\sum x = 5680$

Mean: $\bar{x} = \frac{\sum x}{n}$

$\bar{x} = \frac{5680}{12}$

$\boxed{\bar{x} = 473.33}$

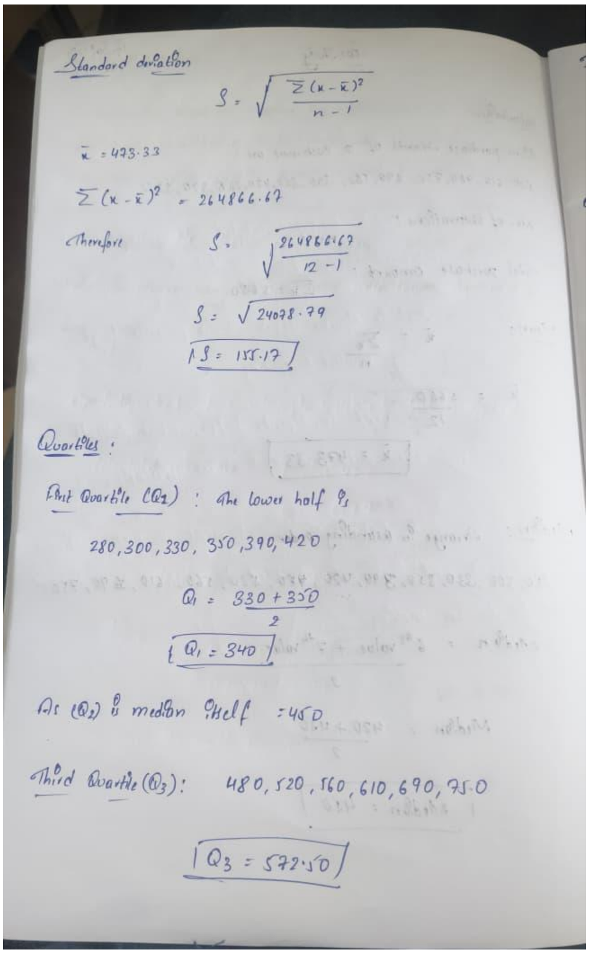
Median: Arrange in ascending order

280, 300, 330, 350, 390, 420, 480, 520, 560, 610, 690, 750

$\text{Median} = \frac{6^{th}\text{ value} + 7^{th}\text{ value}}{2}$

$\text{Median} = \frac{420 + 480}{2}$

$\boxed{\text{Median} = 450}$



Standard deviation

$S = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$

$\bar{x} = 473.33$

$\sum (x-\bar{x})^2 = 264866.67$

Therefore $S = \sqrt{\frac{264866.67}{12-1}}$

$S = \sqrt{24078.79}$

$\boxed{S = 155.17}$

Quartiles :

First Quartile (Q_1) : The lower half is

280, 300, 330, 350, 390, 420

$Q_1 = \frac{330 + 350}{2}$

$\boxed{Q_1 = 340}$

As (Q_2) is median itself $= 450$

Third Quartile (Q_3): 480, 520, 560, 610, 690, 750

$\boxed{Q_3 = 572.50}$

Iterquartile Range (IQR)

$IQR = Q_3 - Q_1$

$IQR = 572.50 - 345$

$\boxed{IQR = 227.50}$

Outlier detection

lower bound $= Q_1 - 1.5(IQR)$

$= 345 - 1.5(227.50)$

$= 3.75$

Upper bound : $Q_3 + 1.5(IQR)$

$= 572.50 + 1.5(227.50)$

$= 913.75$

Therefore, any purchase amount : $x < 3.75$ (or) $x > 913.75$

would be considered outlier

So, No outliers.

Hypothesis Testing

Premium customer purchase Amounts

520, 610, 480, 750, 690, 560

Total : 3610

$n_1 = 6$

Mean : $\bar{x}_1 = \frac{3610}{6}$

$\bar{x}_1 = 601.67$

Regular customer purchase Amounts

300, 350, 420, 280, 390, 330

Total : 2070

$n_2 = 6$

Mean : $\bar{x}_2 = \frac{2070}{6}$

$\bar{x}_2 = 345$

So, $601.67 > 345$

Null Hypothesis (H_0)

$H_0 : \mu_{premium} = \mu_{Regular}$

There is no significant difference

Alternative Hypothesis (H_1)

$H_1 : \mu_{premium} > \mu_{Regular}$

Premium customers spend significantly more than regular customers

T-Test

An independent two-sample welch's t-test is applied

calculated vals are $t = 5.422$

$P = 0.0079$

Significance level : $\alpha = 0.05$

Since $P < 0.05$

$0.00079 < 0.05$

we reject the null hypothesis.

CODE:

```
from flask import Flask
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns
import os
import webbrowser
import threading

app = Flask(__name__)

# CSV file
CSV_FILE = "customer_data.csv"


@app.route("/")
def home():

    df = pd.read_csv(CSV_FILE)

    purchase = df["Purchase Amount"]


    mean = purchase.mean()
    median = purchase.median()
    std = purchase.std()

    q1 = purchase.quantile(0.25)
```

```
q3 = purchase.quantile(0.75)
iqr = q3 - q1

lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

outliers = df[
    (purchase < lower) |
    (purchase > upper)
]
premium = df[
    df["Type"] == "Premium"
]["Purchase Amount"]

regular = df[
    df["Type"] == "Regular"
]["Purchase Amount"]

premium_mean = premium.mean()
regular_mean = regular.mean()
t_stat, p_value = stats.ttest_ind(
    premium,
    regular,
    equal_var=False
)

if p_value < 0.05:
    conclusion = (
        "Reject H0: Premium customers "
        "spend significantly more."
    )
else:
    conclusion = (
        "Fail to reject H0: "
        "No significant difference."
    )
graph_folder = "graphs"

os.makedirs(graph_folder, exist_ok=True)

plt.figure(figsize=(8, 5))

plt.bar(
    df["Customer"],
    df["Purchase Amount"]
)

plt.title("Customer Purchase Amount")
plt.xlabel("Customer")
plt.ylabel("Purchase Amount")
```

```
plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "purchase.png"
    )
)

plt.close()

plt.figure(figsize=(7, 5))

sns.barplot(
    x="Type",
    y="Purchase Amount",
    data=df
)

plt.title(
    "Premium vs Regular Customer Spending"
)

plt.xlabel("Customer Type")
plt.ylabel("Average Purchase Amount")

plt.tight_layout()

plt.savefig(
    os.path.join(
        graph_folder,
        "customer_type.png"
    )
)

plt.close()

plt.figure(figsize=(7, 5))

sns.scatterplot(
    x="Browsing Time",
    y="Purchase Amount",
    data=df
)

plt.title(
    "Browsing Time vs Purchase Amount"
)
```

```
    plt.xlabel("Browsing Time (minutes)")
    plt.ylabel("Purchase Amount")

    plt.tight_layout()

    plt.savefig(
        os.path.join(
            graph_folder,
            "browsing.png"
        )
    )

    plt.close()

    plt.figure(figsize=(7, 5))

    sns.histplot(
        df["Purchase Amount"],
        bins=6,
        kde=True
    )

    plt.title(
        "Distribution of Purchase Amount"
    )

    plt.xlabel("Purchase Amount")
    plt.ylabel("Number of Customers")

    plt.tight_layout()

    plt.savefig(
        os.path.join(
            graph_folder,
            "distribution.png"
        )
    )

    plt.close()
    table = df.to_html(
        index=False,
        classes="data-table"
    )

    html = f"""
<!DOCTYPE html>

<html>

<head>
```

```
<title>E-Commerce Customer Analysis</title>

<style>

body {{
    font-family: Arial, sans-serif;
    background: #f2f4f7;
    margin: 0;
    padding: 30px;
}}

.container {{
    max-width: 1100px;
    margin: auto;
    background: white;
    padding: 30px;
    border-radius: 12px;
}}

h1 {{
    text-align: center;
}}

h2 {{
    margin-top: 35px;
}}

table {{
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
}}

th, td {{
    border: 1px solid #ddd;
    padding: 10px;
    text-align: center;
}}

th {{
    background: #333;
    color: white;
}}

.cards {{
    display: grid;
    grid-template-columns:
        repeat(3, 1fr);
    gap: 15px;
```

```
}}

.card {{
   background: #eef2f5;
   padding: 20px;
   border-radius: 10px;
   text-align: center;
}}

.card h3 {{
   margin: 5px;
}}

.value {{
   font-size: 25px;
   font-weight: bold;
}}

.result {{
   background: #eef6ee;
   padding: 20px;
   border-radius: 10px;
}}

.graphs {{
   display: grid;
   grid-template-columns:
      1fr 1fr;
   gap: 25px;
}}

.graphs img {{
   width: 100%;
   border-radius: 8px;
   border: 1px solid #ddd;
}}

</style>

</head>


<body>

<div class="container">

<h1>
E-Commerce Customer Analysis
</h1>
```

<h2>
Customer Dataset
</h2>

{table}

<h2>
Descriptive Statistics
</h2>

<div class="cards">

<div class="card">

<h3>
Mean
</h3>

<div class="value">
{mean:.2f}
</div>

</div>

<div class="card">

<h3>
Median
</h3>

<div class="value">
{median:.2f}
</div>

</div>

<div class="card">

<h3>
Standard Deviation
</h3>

<div class="value">
{std:.2f}

</div>

</div>

<div class="card">

<h3>
Q1
</h3>

<div class="value">
{q1:.2f}
</div>

</div>

<div class="card">

<h3>
Q3
</h3>

<div class="value">
{q3:.2f}
</div>

</div>

<div class="card">

<h3>
IQR
</h3>

<div class="value">
{iqr:.2f}
</div>

</div>

</div>

<h2>
Outlier Analysis
</h2>

```
<div class="result">

<p>
<b>Lower Bound:</b>
{lower:.2f}
</p>

<p>
<b>Upper Bound:</b>
{upper:.2f}
</p>

<p>
<b>Number of Outliers:</b>
{len(outliers)}
</p>

</div>


<h2>
Premium vs Regular Customers
</h2>


<div class="cards">


<div class="card">

<h3>
Premium Average
</h3>

<div class="value">
{premium_mean:.2f}
</div>

</div>


<div class="card">

<h3>
Regular Average
</h3>

<div class="value">
{regular_mean:.2f}
</div>
```

</div>

</div>

<h2>
Hypothesis Testing
</h2>

<div class="result">

<p>
Null Hypothesis:
Premium and Regular customers
have the same average spending.
</p>

<p>
Alternative Hypothesis:
Premium customers spend more.
</p>

<p>
t-statistic:
{t_stat:.3f}
</p>

<p>
p-value:
{p_value:.5f}
</p>

<p>
Significance Level:
0.05
</p>

<p>
Conclusion:
{conclusion}
</p>

</div>

<h2>
Data Visualization

</h2>

<div class="graphs">

<div>

<h3>
Customer Purchase Amount
</h3>

<img src="/graphs/purchase.png">

</div>

<div>

<h3>
Premium vs Regular Spending
</h3>

<img src="/graphs/customer_type.png">

</div>

<div>

<h3>
Browsing Time vs Purchase Amount
</h3>

<img src="/graphs/browsing.png">

</div>

<div>

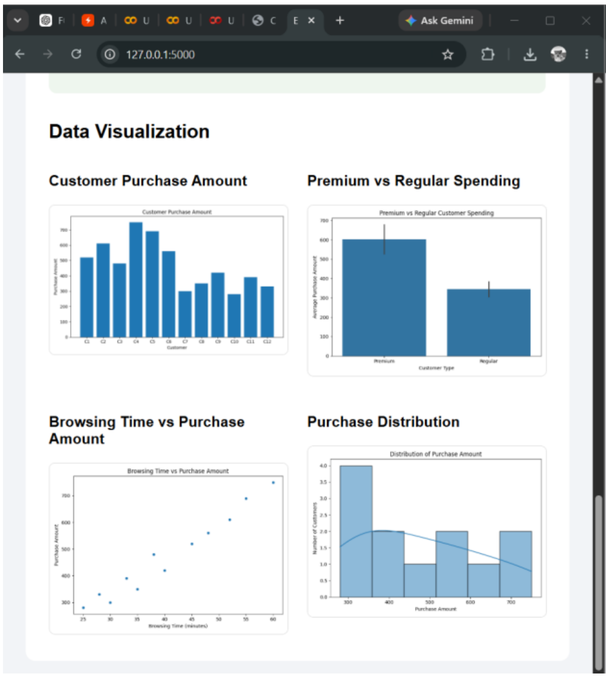
<h3>
Purchase Distribution
</h3>

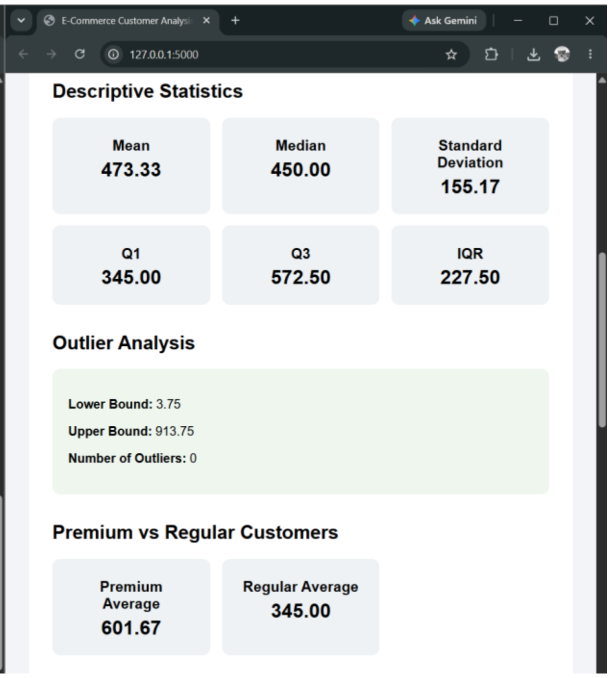
<img src="/graphs/distribution.png">

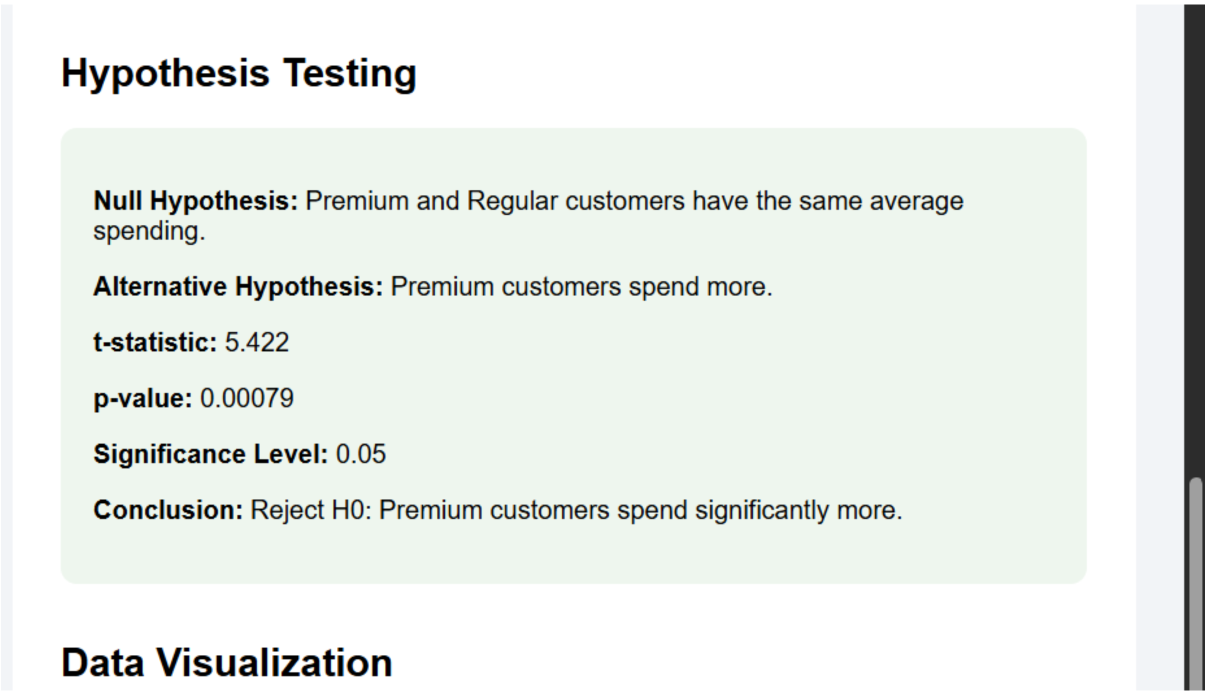
</div>

```
</div>

</div>

</body>

</html>
"""

    return html

@app.route("/graphs/<filename>")
def graphs(filename):

    from flask import send_from_directory

    return send_from_directory(
        "graphs",
        filename
    )

def open_browser():

    webbrowser.open(
        "http://127.0.0.1:5000"
    )

if __name__ == "__main__":

    threading.Timer(
        1.5,
        open_browser
    ).start()

    app.run(
        debug=False,
        port=5000
    )
```

OUTPUT:


127.0.0.1:5000
Data Visualization
Customer Purchase Amount
Premium vs Regular Spending
Browsing Time vs Purchase Amount
Purchase Distribution


E-Commerce Customer Analysis
127.0.0.1:5000
Descriptive Statistics
Mean
473.33
Median
450.00
Standard Deviation
155.17
Q1
345.00
Q3
572.50
IQR
227.50
Outlier Analysis
Lower Bound: 3.75
Upper Bound: 913.75
Number of Outliers: 0
Premium vs Regular Customers
Premium Average
601.67
Regular Average
345.00


Hypothesis Testing
Null Hypothesis: Premium and Regular customers have the same average spending.
Alternative Hypothesis: Premium customers spend more.
t-statistic: 5.422
p-value: 0.00079
Significance Level: 0.05
Conclusion: Reject H0: Premium customers spend significantly more.
Data Visualization